

Cooperative Load Transport with Movable Load Center of Mass Using Multiple Quadrotor UAVs

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Abstract— This paper presents a strategy for cooperative load transport with movable load center of mass by multiple quadrotor UAVs. The load center of mass is modeled by a mass-spring system that moves in the x-y plane in time. The center of mass moving in the x-y plane by a sinusoid functions of time. We use the Newtonians law to derive the systems dynamics equations. The load is connected to the quadrotors by cables. The maximum thrust of the single quadrotor is less than required for load transportation. For this reason we use four quadrotor UAVs for load transportation. The leader-follower structure is applied for cooperate the quadrotors. We use a feedback linearization controller to control quadrotors along the z axis. Finally, some numerical simulations are considered to validate the proposed method.

Keywords—cooperative load transportation, movable load center of mass, quadrotor UAVs, leader-follower structure, feedback linearization controller

I. INTRODUCTION

Control of multi-agent systems is the new field in control engineering that inspired by biological behavior such as flocking of birds and schooling of the fishes. In the last years, researchers have more focused on controlling of multi-agent systems. One of the methods, for controlling multi-agent systems is formation control. This method, is more suitable for controlling different type of robots such as mobile robots, unmanned aerial vehicles, and autonomous underwater vehicles. Unmanned aerial vehicles (UAVs) have found more applications for research, transportation, mapping and so on [1].

The main structures that considered for the formation control of multi-vehicle systems are leader-follower, virtual, and behavioral structures. In the leader-follower structure, one of the agents is the leader and the other agents follow it. In this structure, the followers track the leader and there is no feedback from the followers to the leader. For this reason, if the leader fails, the mission will fail too and if each of followers fails, there is no guarantee that the formation is kept. But this structure is easy to understand and it is not difficult to implement [2], [3]. In the virtual structure, the leader is a virtual leader and usually is a virtual point. The main advantage

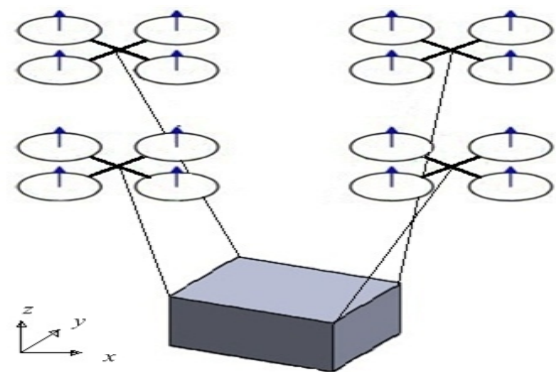


Fig. 1. Load transportation with multiple quadrotor.

of this structure is that there is no fail for the leader, and the main disadvantage of this structure, is that it cannot modify the formation [4], [5]. In the behavioral structure, behavior of each agent is defined according to the purpose of system such as obstacle avoidance or collision or other issues [6].

Quadrotor UAVs have been used in a wide range of applications and becomes more useful for mobile platforms in academic fields; especially researchers are more interested in load transporting via quadrotor. In [7], the problem of carrying slung load was considered, and an adaptive control method for the quadrotors to carrying a slung load was addressed. In this paper, the slung load modeled by a pendulum mass that hanged from a point. There existed a lot of works about load transportation with single quadrotor, however if the load was heavy, a single quadrotor could not transport the load. In [8], multiple quadrotors with cooperatively grasp and transport a payload with different shapes for payload was studied. A gripper that attached to the payload was used and different shapes of the payload were considered. In [9], the problem of controlling a team of the quadrotors with manipulating and transporting a payload with cables was investigated. Methods were tested on a team of quadrotors. In [10], the problem of tracking a trajectory for an arbitrary number of quadrotors to transport a load was addressed. In that paper, was assumed that the load is a mass point and connected to the UAVs by the

rigid massless links. In [11], a method for tracking control for a rigid body payload that is connected to an arbitrary number of quadrotors was considered (see Fig. 1). The rigid body payload was connected to the UAVs by rigid links. In [12], the problem of cooperative transportation of a cable-suspended payload by multiple quadrotors with complete dynamic of payload was studied.

In all past studies, the center of mass of payload was considered fixed. However, in this work, we consider the load with movable center of mass that changes its position by time. We model this problem by a mass-spring system that vibrate in two dimensions, x and y . The load is connected to the quadrotors by cables and the agents cooperate with each other in a leader-follower structure.

The rest of the paper is organized as follows. In Section 2, we describe the quadrotor dynamics. In Section 3, the leader-follower structure is considered. In Section 4, the problem is described. A proper controller is developed for the system, formulated in section 5, and In Section 6, we validate our method by a numerical example in Matlab.

II. QUADROTOR DYNAMICS

Quadrotor are modeled by Euler-Lagrange method. Using Lagrange formula the equations of motion are obtained as:

$$\begin{aligned} I_{xx}\ddot{\phi} &= \dot{\theta}\dot{\psi}(I_{yy} - I_{zz}) \\ I_{yy}\ddot{\theta} &= \dot{\phi}\dot{\psi}(I_{yy} - I_{zz}) \\ I_{zz}\ddot{\psi} &= \dot{\theta}\dot{\phi}(I_{xx} - I_{yy}) \\ m\ddot{z} &= (c\psi c\phi) \sum T_i \\ m\ddot{x} &= (s\psi s\phi + c\psi s\theta c\phi) \sum T_i \\ m\ddot{y} &= (s\psi s\theta c\phi - c\psi s\phi) \sum T_i \end{aligned} \quad (1)$$

where c is $\cos(\cdot)$ and s is $\sin(\cdot)$. ϕ, θ and ψ are pitch, roll and yaw angular. I_{xx} is the moment of inertia around the x axis, I_{yy} is the moment of inertia around the y axis and I_{zz} is the moment of inertia around the z axis. T_i is the torque that produced by rotor i . We use equation (1) to write the system in state-space form $\dot{X} = f(X, U)$ with U as the input vector and X as the state vector. The State vector is:

$$X = [\phi \ \theta \ \psi \ \dot{\phi} \ \dot{\theta} \ \dot{\psi} \ z \ \dot{z} \ x \ \dot{x} \ y \ \dot{y}]^T \quad (2)$$

The input vectore consists of:

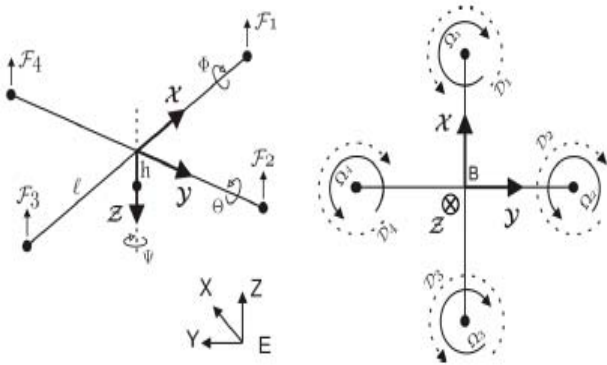


Fig. 2. Quadrotor coordinate system [13].

$$U = [u_1 \ u_2 \ u_3 \ u_4]^T \quad (3)$$

where

$$\begin{aligned} u_1 &= b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2) \\ u_2 &= b(-\Omega_2^2 + \Omega_4^2) \\ u_3 &= b(\Omega_1^2 - \Omega_3^2) \\ u_4 &= d(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2) \end{aligned} \quad (4)$$

where b is the thrust coefficient, d is the drag coefficient, and Ω_i is the angular velocity of rotor i .

Finally, we obtain [13]:

$$f(X, U) = \begin{pmatrix} \dot{\phi} \\ \dot{\theta}\dot{\psi}a_1 + \dot{\theta}a_2\Omega_r + b_1U_2 \\ \dot{\theta} \\ \dot{\phi}\dot{\psi}a_3 - \dot{\phi}a_4\Omega_r + b_2U_3 \\ \dot{\psi} \\ \dot{\theta}\dot{\phi}a_5 + b_3U_4 \\ \dot{z} \\ g - (\cos \phi \cos \theta) \frac{1}{m}U_1 \\ \dot{x} \\ u_x \frac{1}{m}U_1 \\ \dot{y} \\ u_y \frac{1}{m}U_1 \end{pmatrix} \quad (5)$$

with:

$$\begin{aligned} a_1 &= (I_{yy} - I_{zz})/I_{xx} \\ a_2 &= J_r/I_{xx} \\ a_3 &= (I_{zz} - I_{xx})/I_{yy} \\ a_4 &= J_r/I_{yy} \\ a_5 &= (I_{xx} - I_{yy})/I_{zz} \end{aligned} \quad (6)$$

J_r is rotational moment of inertia.

$$\begin{cases} b_1 = l/I_{xx} \\ b_2 = l/I_{xx} \\ b_3 = 1/I_{xx} \end{cases} \quad (7)$$

$$\begin{cases} u_x = (\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \\ u_y = (\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \end{cases} \quad (8)$$

and

$$\Omega_r = -\Omega_1 + \Omega_2 - \Omega_3 + \Omega_4 \quad (9)$$

III. LEADER FOLLOWER STRUCTURE

In this section, we consider the leader-follower structure for multi agent systems. This structure is the most popular structure in formation control. In this structure, one or some agents are considered as leader or leaders and the others follow it or them. The leader tracks a desired given trajectory and the followers keep the desired relative position and orientation to the leader. This method is easy to understand and implement but there is some disadvantages such as:

1. Because there is no feedback from followers to the leader, any fault in followers may fail the mission.

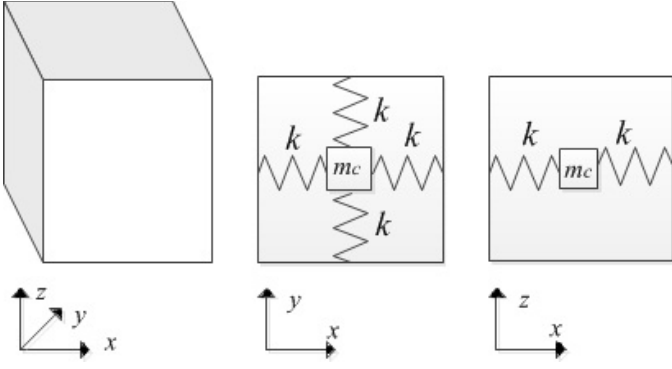


Fig. 3. Load model.

2. Because the leader is the target for the followers, if the leader fails, the whole mission will fail.

However, using more than one leader can be a solution for the second disadvantage.

Knowing the fact that having a desired trajectory defined for the leader will be good enough to have the followers to keep the relative desired position the following controller for each follower is defined:

$$U_i = k_1(q_l - q_i - d_i) + k_2(\dot{q}_l - \dot{q}_i) \quad (10)$$

where q_l is the leader's state, q_i is the state of follower i , d_i is the desired relative position of follower i from the leader, k_1 and k_2 are positive constants. [2], [3].

IV. LOAD TRANSPORTATION MODEL

A. modeling of movable center of mass

In this section, we model the load with movable center of mass, by a mass and four springs that is shown in Fig. 3. The mass position can change by time. The Mass-spring system can move in the x-y plane. We derive the equations of motion by Newtonian laws. The equations of motion for the mass-spring system are:

$$\begin{aligned} F_x - k\Delta_x &= m_c a_x \\ F_y - k\Delta_y &= m_c a_y \\ F_z - m_c g &= m_c a_z \end{aligned} \quad (11)$$

where F_x , F_y and F_z are the forces along the x, y and z axes that apply from the quadrotors, respectively. Δ_x and Δ_y are the displacements of the springs along the x and y axes. a_x , a_y and a_z are the acceleration of mass along the x, y and z axes. m_c is the mass of system.

B. The overall system modeling

In this section, we add dynamics of the mass-spring system to the quadrotors dynamics.

To do so, the following assumptions are considered:

1. The quadrotors are equal in size and power and they are symmetric with respect to the load.

2. The load system is a box with a mass point and four springs that mass can move along the x and y axes.
3. The box, springs and the cables are massless.
4. The cables do not stretch.
5. The load acceleration is equal to the average of the quadrotors acceleration.
6. The cables are attached to the center of mass of the quadrotors.
7. The coefficient stiffness of springs are equal.

Now, we can rewrite the state space model of quadrotors with load as follows. Note that the effect of load is applied along the x, y and z axes and F_x , F_y and F_z of the mass-spring system add to the equation of the quadrotors along the x, y and z axes.

$$f(X, U) = \begin{pmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \\ \dot{\varphi} a_1 + \dot{\theta} a_2 \Omega_r + b_1 U_2 \\ \dot{\theta} \\ \dot{\psi} \\ \dot{\varphi} a_3 - \dot{\theta} a_4 \Omega_r + b_2 U_3 \\ \dot{\psi} \\ \dot{\theta} \varphi a_5 + b_3 U_4 \\ \dot{z} \\ g - (\cos \varphi \cos \theta) \frac{1}{m} U_1 + \frac{m_c g}{nm} \\ u_x \frac{1}{m} U_1 + \frac{k \Delta_x}{n} - \frac{a_x m_c}{n} \\ \dot{y} \\ u_y \frac{1}{m} U_1 + \frac{k \Delta_y}{n} - \frac{a_y m_c}{n} \end{pmatrix} \quad (12)$$

where n is the number of quadrotors, m_c is the mass of load, k is the stiffness of springs, Δ_x is the displacement of springs along the x axis, Δ_y is the displacement of springs along the y axis, a_x is the acceleration of load along the x axis and a_y is the acceleration of load along the y axis.

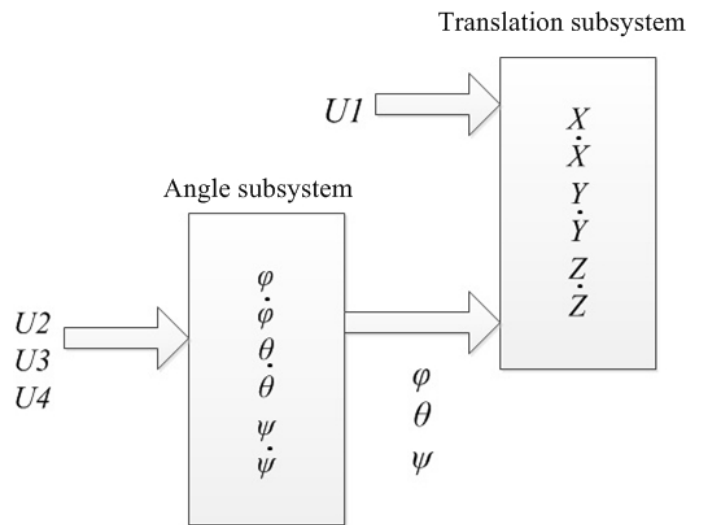


Fig. 4. Angle and translation subsystems.

V. CONTROLLER

It's seen from equation (5) that the angles and their rates are independent from translation components. But the translations depend on the angles. We take system as two subsystems, the angular rotations and the linear translations (see Fig. 4).

A. controller for angular positions

For angular positions, the following P-D controller is used [14]:

$$U_2 = -\frac{l_x}{l} e_\varphi - k_1 \dot{\varphi} \quad (13)$$

$$U_3 = -\frac{l_y}{l} e_\theta - k_2 \dot{\theta} \quad (14)$$

$$U_4 = -l_z e_\psi - k_3 \dot{\psi} \quad (15)$$

where e_φ is the difference between φ and φ_{desire} , e_θ is the difference between θ and θ_{desire} , e_ψ is the difference between ψ and ψ_{desire} , l is the distance between each rotors to the quadrotors center of mass. k_1 , k_2 and k_3 are positive constants.

B. Altitude controller

For altitude, a feedback linearization controller is proposed:

$$U_1 = \frac{m}{(\cos \theta \cos \varphi)} (g + \frac{m_c g}{nm} + e_z - k_4 e_{\dot{z}} - k_5 e_z) \quad (16)$$

where:

$$\begin{cases} e_z = z - z_{desire} \\ e_{\dot{z}} = \dot{z} - \dot{z}_{desire} \\ e_z = \ddot{z} - \ddot{z}_{desire} \end{cases} \quad (17)$$

and k_4 and k_5 are positive constants, and n is the number of quadrotors. m is the mass of quadrotor and m_c is the mass of load.

VI. SIMULATION AND RESULTS

In this section we consider a numerical simulation in Matlab. We use 4 quadrotors. The initial positions for leader is (0,0,1), and followers are (-1,0,1), (0,1,1), (-1,1,1). The mass of each quadrotor is 1kg. The mass of load is 2 kg. The stiffness of each spring is 5N/m, so total stiffness along x and y is 10 N/m. we suppose that mass vibrate by $\Delta_{x,y} = 0.1 \sin(t)$. The distances between quadrotors are 1m that is the length of load. It is our goal to have the leader position (6,7,8), so the followers positions should be at (5,7,8), (6,8,8), and (5,8,8). In this condition, the trajectories of the quadrotors are depicted in Figures 5, 6, 7, and 8 confirming the accuracy of the proposed strategy.

VII. CONCLUSION

This paper, considers the problem of load transportation with multiple quadrotors. The load that was considered is a box with movable center of mass. The center of mass vibrates in the x-y plane. A leader-follower structure is considered to achieve a formation. We are now working on some other structures for quadrotors cooperation such as virtual leader.

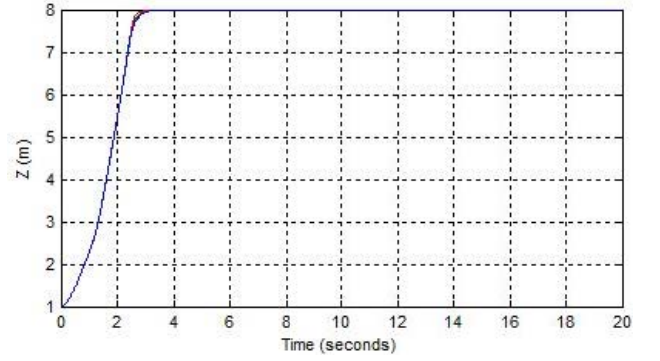


Fig. 5. Heights of quadrotors.

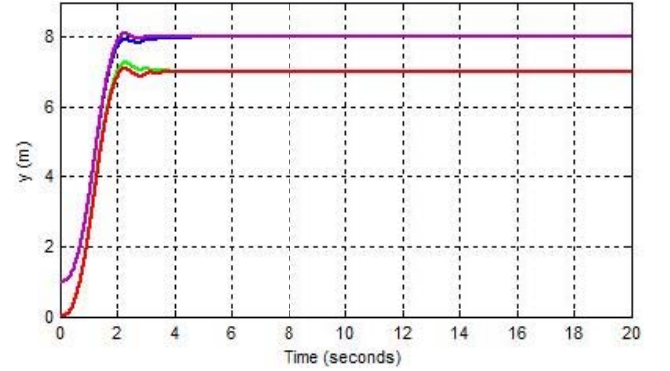


Fig. 6. Widths of quadrotors.

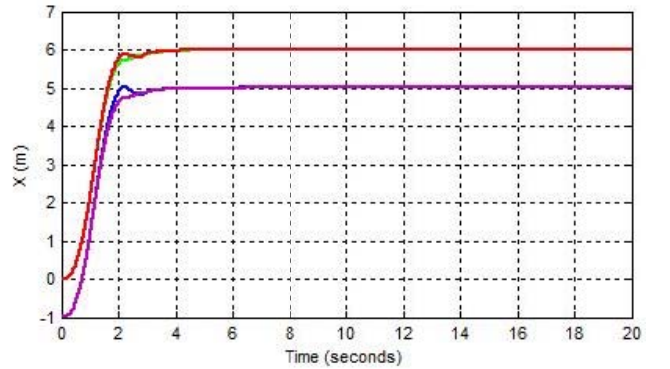


Fig. 7. Length of quadrotors.

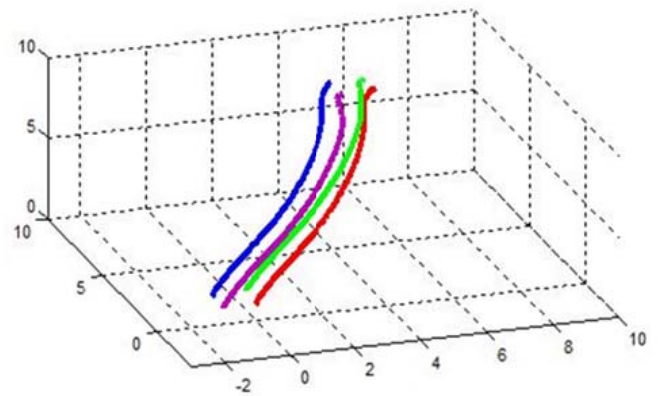


Fig. 8. 3D trajectories of quadrotors.

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